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Title: HFNC Algorithim Use Outside of PICU

High Flow Nasal Cannula(HFNC) Initiation Algorithm

	2. Anticipated Intolerance to mask or interface Common diseases for HFNC use ❑ Bronchiolitis most common indication ❑ Other respiratory diseases causing respiratory distress such as pneumonia, asthma
Contra-indications	❑ Significant respiratory distress(relative) / respiratory failure ❑ Hemodynamic instability ❑ Acute respiratory Distress Syndrome with significant hypoxemia ❑ Encephalopathy

Settings	Infant : Start Flow at 2l/kg/min,Fio2 60% Child : Start Flow 1 l/kg/min , Fio2 60% ❑ Aim for saturation 93%-94% ❑ Titrate Fio2 to clinical response

Monitoring	❑ Improvement in the work of breathing and accessory muscles use ❑ Improvement in oxygen saturation and blood gas ❑ Frequent clinical assessment required ❑ Most achieve clinical response within first 60 -90 minutes

Improvement in work of breathing, saturation and blood gas

- ❑ Continue clinical monitoring
- ❑ Consider feeding via NGT feeds within 4 hours if child is stable

No improvement in the work of breathing, saturation or blood gas

- ❑ Consult PICU
- ❑ Escalate treatment to invasive ventilation if in respiratory failure or non-invasive ventilation if not in respiratory failure
- ❑ Go to non-invasive ventilation algorithm

Wean Fio2 if saturation is > 94% following first 2 hours of stabilization

Wean Fio2 to 40% gradually as able

improvement In respiratory distress

Use Respiratory assessment score(RAS)

Reduce Flow by 50% every 12 hours if no respiratory distress

Assess RAS 20 minutes from the flow change

Respiratory Assessment Score

Age Variable	Respiratory Rate		
	No/mild	Moderate	Severe
< 2 mos	</= 60	61 - 69	>/= 70
2 mos – 1 yr	</= 50	51 - 59	>/= 60
1 – 2 yr	</= 40	41 - 44	>/= 45
2 – 3 yr	</= 34	35 - 39	>/= 40
4 – 5 yr	</= 30	31 - 35	>/= 36
6 – 12 yr	</= 26	27 - 30	>/= 31
>12 yr	</= 23	24 - 27	>/= 28
Intercostal Retraction	none	minimal	marked
Xiphoid Retraction	none	minimal	marked
Nasal Flaring	none	minimal	marked
Expiratory Grunt	none	audible with stethoscope	audible

Remained free of distress

- Convert to low flow oxygen if flow rate < 1l/kg/min
- Continue on low flow oxygen and monitor for respiratory distress
- Consider transfer out if remains free of distress for 8-12 hours

Moderate respiratory distress
Go to previous flow and reassess weaning in 12-24 hours

Significant respiratory distress

- Stay on initial flow rates
- Assess need to escalate to NIV
- Consider slow weaning of 1 l every 4-6 hours in those with difficult to wean

Suggested low flow oxygen rates

Age	Flow Rate (L/min)
< 2 months	0.5
2 months – 2 years	1-1.5
3 – 4 years	2-3
> 4 years	No more than 4

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